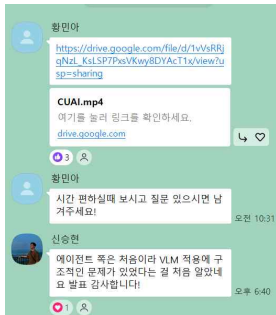


# CUAI 딥러닝 논문 리뷰 스터디 (MM)

2025.11.18

발표자 : 이채린

## 스터디원 소개 및 만남 인증



스터디원 1 : 김현수

스터디원 2 : 신승현

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# 🎵 Music's Multimodal Complexity in AVQA: Why We Need More than General Multimodal LLMs

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## Abstract



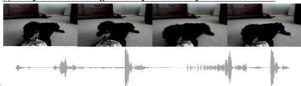
While recent Multimodal Large Language Models exhibit impressive capabilities for general multimodal tasks, specialized domains like music necessitate tailored approaches. Music Audio-Visual Question Answering (Music AVQA) particularly underscores this, presenting unique challenges with its continuous, densely layered audio-visual content, intricate temporal dynamics, and the critical need for domain-specific knowledge. Through a systematic analysis of Music AVQA datasets and methods, this position paper identifies that **specialized input processing, architectures incorporating dedicated spatial-temporal designs, and music-specific modeling strategies are critical for success in this domain.** Our study provides valuable insights for researchers by highlighting effective design patterns empirically linked to strong performance, proposing concrete future directions for incorporating musical priors, and aiming to establish a robust foundation for advancing multimodal musical understanding. This work is intended to inspire broader attention and further research, supported by a continuously updated anonymous GitHub repository of relevant papers: <https://github.com/xid32/Survey4MusicAVQA>.

## 배경

Music AVQA

- 밀집되고 계층화된 신호 해석 (Dense and Layered Signals)
  - 정밀한 시공간적 추론 (Precise Temporal and Spatial Reasoning)
  - 도메인 특정 지식 (Domain-Specific Knowledge)
  - 공간-시간적 상관관계(spatial-temporal correspondence)
- MLLM의 한계

*(i) Temporal Reasoning under Sparse Audio from VGG-Sound*



What does the black puppy do after barking suddenly?



It **bites the pillow** shortly after the sudden barking.

*(ii) Rhythm Comparison under Dense Audio from Music-AVQA*



Is the guzheng on the right more rhythmic than acoustic guitar?

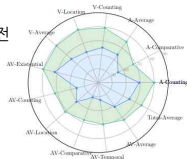


**Yes**, guzheng produces more distinct and repeated plucking patterns.

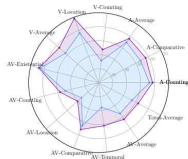
## Music AVQA Datasets

| 데이터셋            | 특징                           | 한계                      |
|-----------------|------------------------------|-------------------------|
| MUSIC-AVQA      | 최초 대규모 음악 AVQA 벤치마크          | 악기 편향, 일부 질문 패턴 단순      |
| MUSIC-AVQA v2.0 | 질문 템플릿 밸런싱 → 답변 편향 완화        | 여전히 장르 다양성 제한           |
| MUSIC-AVQA-R    | 질문 표현 다양화, 분포 이동 → 모델 일반화 평가 | 일부 복잡한 질문/시간적 패턴 처리 어려움 |

→ 질문/답변 편향을 줄이고,  
일반화 능력을 평가하기 위한 구조로 발전



(a) Methods on Music-AVQA [69].



(b) Methods on Music-AVQA-R [86].

## Music AVQA

- LAST-ATT  
transformer + Attention  
모달 간 cross-attention  
음악도메인 부족

- PSTP-Net  
CNN기반  
시각+오디오 late fusion  
시계열, 이벤트 분리 취약

| METHOD              | Text Encoder          | Visual Encoder            | Audio Encoder                 | S-T |
|---------------------|-----------------------|---------------------------|-------------------------------|-----|
| AMUSE [24]          | Transformer [105]     | Swin-Transformer-v2 [81]  | HTS-AT [10]                   | ✓   |
| AUDIO FLAMINGO [61] | OPT-IML-MAX-1.3B [54] | -                         | ClapCap [30]                  | ✓   |
| AVMoE [14]          | -                     | Swin-Transformer-v2 [81]  | HTS-AT [10]                   | ×   |
| AVSD [96]           | LSTM                  | LSTM                      | LSTM                          | ×   |
| AVSIAM [75]         | -                     | ViT [93]                  | ViT [93]                      | ×   |
| AVST [69]           | LSTM                  | ResNet-18 [46]            | VGGish [37]                   | ✓   |
| CAT [119]           | LLaMA2-7B [103]       | ImageBind [38]            | ImageBind [38]                | ×   |
| CHATBRIDGE [130]    | Vicuna-13B [16]       | Vit-G [101]               | BEATs [11]                    | ×   |
| CIGN [88]           | -                     | ResNet-18 [46]            | ResNet-18 [46]                | ✓   |
| COCA [63]           | Word Embedding        | ResNet-18 [46]            | VGGish [37]                   | ×   |
| CONVLSTM [32]       | LSTM                  | -                         | Conv                          | ×   |
| CROSSMAE [42]       | -                     | MAE [45]                  | AudioMAE [51]                 | ×   |
| DCL [84]            | DeBERTa-V3-Large [47] | ViT [93]                  | AST [39]                      | ✓   |
| DG-SCT [27]         | -                     | ViT [93]                  | HTS-AT [10]                   | ✓   |
| EEMC [108]          | RoBERTa [133]         | ViT [93]                  | VGGish [37]                   | ✓   |
| FCNLSTM [32]        | LSTM                  | -                         | Conv                          | ×   |
| GPT-4o [53]         | Transformer           | CLIP-ViT                  | Transformer                   | ×   |
| GRU [4]             | LSTM                  | VGGNet [99]               | -                             | ×   |
| HCRN [65]           | BiLSTM                | ResNet-18 [46]            | -                             | ×   |
| LAST-ATT [80]       | LSTM                  | Swin-Transformer-v2 [81]  | Audio-Spectrogram-Transformer | ✓   |
| LAVISH [76]         | -                     | ViT [93]                  | ViT [93]                      | ✓   |
| LAViT [123]         | Transformer [105]     | Transformer [105]         | Transformer [105]             | ✓   |
| LSTTA [77]          | CLIP [93]             | CLIP [93]                 | w2v-Conformer [41]            | ✓   |
| MAVEN [85]          | Mixtral               | InternViT-300M-448px [13] | Transformer                   | ×   |
| MCAN [121]          | GloVe [90]+LSTM       | Faster R-CNN [94]         | -                             | ×   |
| MCCD [86]           | -                     | -                         | -                             | ✓   |
| MEERKAT [17]        | LLaMA2-7B [103]       | CLIP-ViT                  | CLAP [29]                     | ✓   |
| OGM [110]           | -                     | ResNet-18 [46]            | ResNet-18 [46]                | ×   |
| ONELLM [43]         | LLaMA2-7B [103]       | CLIP-ViT                  | Unified Multimodal Encoder    | ×   |
| OPM [110]           | -                     | ResNet-18 [46]            | ResNet-18 [46]                | ×   |
| PSAC [71]           | Word Embedding        | CNN                       | -                             | ×   |
| PSTP-Net [68]       | CLIP [93]             | CLIP [93]                 | VGGish [37]                   | ✓   |
| QAP [74]            | DeBERTa-V2-XLarge     | CLIP [93]                 | CLAP [29]                     | ×   |
| QWEN2.5-VL [5]      | MRoPE [5]             | ViT [93]                  | -                             | ×   |
| REFATOMNET [89]     | BERT                  | ViT [93]                  | -                             | ✓   |
| VALOR [78]          | BERT                  | CLIP [93]                 | AST [39]                      | ×   |
| VAST [12]           | BERT [21]             | ViT [26]                  | BEATs [11]                    | ×   |
| VIDEOLLAMA-2 [15]   | Transformer           | CLIP [93]                 | BEATs [11]                    | ✓   |
| VITA [34]           | Mixtral [58]          | InternViT-300M-448px [13] | CNN                           | ×   |

## 제안된 방법

| 1. 특화된 입력 처리                                                                                                                                                                                                                       | 2. 공간-시간 설계                                                                                                                                                                                             | 3. 음악 특화 모델링                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- 고해상도 비디오 + 고샘플링 음성 입력</li> <li>- 음악 특화 전처리: beat segmentation, Harmonic-Percussive Separation</li> <li>- 오디오 토크나이제이션: 멜 스펙트로그램 패치 기반 처리</li> <li>- 시각 입력: ViT, Swin 등 패치 기반 처리</li> </ul> | <ul style="list-style-type: none"> <li>- 질문 관련 시간 구간 인식 → 모델이 중요한 시점 집중</li> <li>- 공간적 주의 집중 → 소리에 대응하는 시각적 영역 식별</li> <li>- 모달 융합: 단순 late fusion 대신 cross-attention, temporal alignment 사용</li> </ul> | <ul style="list-style-type: none"> <li>- 음악 이벤트 단서 활용: 웨이브폼 피크, MFCC, 비트 추적, 피치 추정</li> <li>- 중간 수준 음악 구조: 템포, 키, 다운비트, 코드 진행 등</li> <li>- 잠재적 음악적 추론 내부적으로 인과관계 추론</li> <li>- 체인 오브 소트reasoning: 중간 추론 단계 학습 → 모델이 "이렇게 생각했다" 설명 가능</li> </ul> |